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Mitigating pore defects in Mo-Re heat pipe joints via longitudinal laser beam oscillation: decoupling and integrated assessment of frequency and amplitude

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Article

Outline

Highlights

- Longitudinal oscillation cannot change pore spatial or size distribution in Mo-10Re joints.
- Oscillation frequency has a significantly stronger effect on porosity than on the maximum pore size.
- Longitudinal oscillation elongates the molten pool and smooths the temperature gradient.
- The Re threshold for acceptable porosity is 206.45, higher frequency causes higher porosity at the same oscillation $V_{average}$

Highlights

Abstract

Graphical abstract

Keywords

1. Introduction

2. Materials and experimental methods

3. Results and discussion

4. Conclusion

Abstract

As a candidate material for structural components in nuclear reactors, molybdenum (Mo) alloys require strict control of weld pore defects. This study employs longitudinal laser beam oscillation welding to reduce pore defects in Mo alloy pipe weld joints and investigates the underlying mechanisms of pore formation and elimination. Computerized tomography (CT) scan reveals that both porosity and maximum pore size decrease with increasing oscillation amplitude, while with increasing oscillation frequency, they first decrease and then increase. At the same oscillation speed, higher frequencies tend to lead to higher porosity. The number of pores initially increases and then decreases along the wall thickness direction, with most pores concentrated near the fusion line. Adjusting oscillation parameters cannot change the spatial or size distribution of pores, which conforms with the characteristics of base metal (BM) release-type pores induced by oxygen. Bayesian analysis reveals that the improvement in porosity due to oscillation frequency is more significant than its effect on reducing maximum pore size, while the effects of oscillation amplitude on both are likely comparable. High-speed photographs demonstrate that longitudinal oscillation elongates the molten pool, making the temperature gradient smoother, but excessively high frequencies can lead to molten pool instability. When the Reynolds number exceeds approximately 206.45, qualified weld joints with a porosity below 1% can be achieved. This study provides theoretical guidance for the efficient control of pore defects in Mo alloy welding and deepens the understanding of the relevant physical processes.

Graphical abstract

CRediT authorship contribution statement

Declaration of competing interest

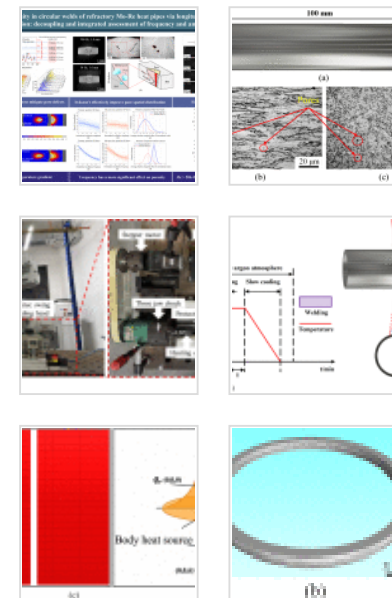
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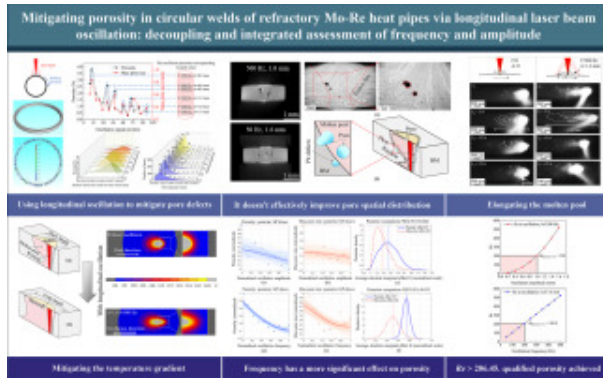


Table 3. Experimental parameters for th...

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Supplementary material

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Keywords

Molybdenum alloy heat pipe; Laser welding; Oscillation; Pore

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1. Introduction

Molybdenum (Mo) and its alloys are promising candidates for advanced nuclear heat-pipe reactors, owing to their excellent high-temperature strength [1], corrosion resistance, low thermal expansion coefficient [2], and small neutron absorption cross-section [3]. While electron beam welding (EBW) is the most mature joining method [4], laser welding (LW) has attracted increasing attention due to its lower cost and higher flexibility [5]. However, researchers [6], [7] have reported that LW of Mo alloys tends to produce serious porosity. Since Mo alloy heat pipes operate above 1000°C with internal pressure, pores cause local stress concentration, significantly reducing sealing

performance and creep resistance, thereby accelerating failure [8]. Therefore, porosity level and maximum pore size, as critical evaluation indicators, must be strictly controlled.

Effective control of pore defects in Mo alloy welds requires clarification of pore formation mechanisms. The manufacturing process significantly influences pore formation. Dong et al. [9] related pores to microvoids and high oxygen content inherent in powder metallurgy (PM) base metal (BM). Zhang et al. [10] noted that residual hydrogen from H₂ shielding during PM processing can be released during welding, promoting pore formation. Metallurgical reactions also contribute to porosity. Wang et al. [11] reported that intragranular pores originate from gas release of pre-existing micropores, while intergranular pores are associated with volatilization of low-boiling-point oxides such as MoO₂ and MoO₃. For Titanium-Zirconium-Molybdenum alloy [12], CO generated in the molten pool cannot escape promptly due to its low solubility and rapid solidification, leading to pore formation. Collectively, pores in Mo alloy welds mainly stem from base-metal-released gas (from PM raw material) and metallurgical gas generated during welding.

Due to cost and technical constraints, most Mo alloy components are fabricated via PM [13]. Without optimizing the BM, pore defects control through welding process adjustment is critical. Existing methods (heat input control [14], laser remelting [10], alloying deoxidation [15]) can reduce porosity but have clear limitations: Heat input adjustment is restricted by weld depth and grain coarsening. Remelting promotes pore escape but causes serious grain growth and performance degradation [16]. Alloying is not allowed in nuclear applications due to strict purity requirements. Thus, conventional strategies have limited engineering value. In contrast, vacuum LW [17], magnetic field-assisted LW [18], and beam oscillation welding [19] all exhibit superior porosity suppression performance.

Among these, laser beam oscillation welding is outstanding for dynamically regulating heat distribution and molten pool behavior to efficiently promote pore escape. Studies [20], [21] show that high-frequency oscillation generates directional flow in the melt

pool, which stabilizes the molten pool and shortens bubble escape paths. The mechanism includes two aspects: periodic stirring drives pore migration; the dynamic keyhole provides a directed channel for pore overflow. Gao et al. [22] further confirmed that pores near the keyhole can be removed by metal vapor flow, while surface pores escape via Marangoni convection. This multi-path synergistic effect greatly reduces porosity. Beam oscillation controls pores by physical means without extrinsic elements, while improving microstructure and mechanical properties [23], which has been widely verified. Kou et al. [24], [25] used arc oscillation to reduce porosity in 2319 and 2319/5356 aluminum (Al) alloy components, enhancing microstructural homogeneity and mechanical performance.

However, most studies focus only on overall porosity and neglect the spatial distribution of pores. Clarifying the detailed pore distribution is critical for further defect elimination. In addition, research on beam oscillation for porosity control in Mo alloy welding remains limited, as most work concentrates on Al alloys and steels. Given the distinct thermophysical properties of Mo alloys, the effects of oscillation parameters, like oscillation mode, require specific investigation.

Longitudinal oscillation is rarely adopted for Al alloys and steels owing to melt pool instability and spattering [26], [27]. However, Mo alloys possess much higher melt viscosity than Al [28], which can effectively stabilize the molten pool. Besides, longitudinal oscillation produces a narrower bead at the same penetration [27]. Combined with the base-metal-released pore mechanism in Mo alloys, a smaller molten volume reduces the total pore formation. Meanwhile, the reciprocating movement acts as localized remelting and facilitates pore escape. These factors indicate that longitudinal oscillation has great potential for porosity control in Mo alloy welding.

Nevertheless, trial-and-error parameter optimization remains time-consuming and inefficient even with a determined oscillation mode. Data-driven modeling has been proven efficient for process optimization, yet high-precision models usually require large datasets. To reduce experimental cost, multi-physics simulation/detection has been combined with intelligent algorithms. Lv et al. [29] developed a multi-physics finite

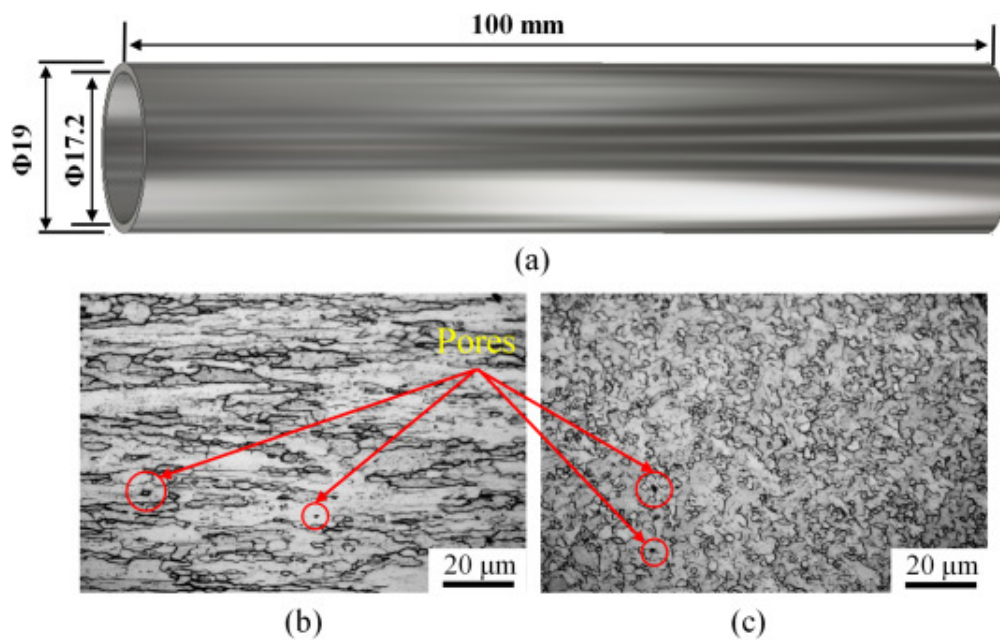
element model for resistance spot welding to generate high-quality synthetic data, supporting efficient machine learning training. Wang et al. [30] proposed a framework integrating molecular dynamics, machine learning, and high-throughput optimization, realizing efficient process optimization for femtosecond laser drilling under small-sample conditions. The Bayesian method is also suitable for small-sample optimization. Haas et al. [31] combined Bayesian optimization with X-ray imaging and height measurement, achieving optimal LW parameters for AA5754 Al alloy using only 29 experiments. Model performance differs significantly for different targets. Cheng et al. [32] compared response surface methodology (RSM) and random forest (RF) in Inconel 718 micro-hole machining: RF was more accurate for material removal rate, while RSM was better for geometric features. Robustness can be improved by expanding the parameter range and using hybrid models. Incorporating physical prior knowledge further improves prediction reliability. Xia et al. [33] embedded physical mechanisms of resistance spot welding into neural networks, developing a physics-informed neural network with significantly enhanced out-of-distribution generalization. Previous studies revealed that adopting suitable data-driven models for specific manufacturing processes, or applying scientific mathematical statistical analysis, enables efficient process optimization.

Based on the aforementioned research background, this study employs X-ray imaging for quantitative characterization of weld pore defects and systematically investigates the effects of longitudinal laser oscillation parameters—frequency and amplitude—on porosity, maximum pore size, pore count, and spatial distribution in Mo-Re alloy pipe joints. A small-sample Bayesian analysis is used to reveal the differential significance of oscillation parameters on porosity versus maximum pore size, identifying oscillation frequency as the efficient parameter for effective pore suppression. High-speed imaging and thermal field simulations further elucidate the underlying mechanisms. Finally, amplitude and frequency are coupled into an average oscillation velocity, and a potential rapid strategy for controlling porosity in Mo alloys is theoretically proposed based on the Reynolds number.

2. Materials and experimental methods

2.1. Materials

The experimental material for the circular weld in this study was Mo-10Re alloy, and the pipe dimensions are shown in Fig. 1(a). The Mo alloy pipes had an outer diameter of 19mm and a wall thickness of about 0.9mm. Fig. 1(b) and (c) present the metallographic structures of the axial and radial cross-sections of the Mo-10Re pipe, respectively. It is noteworthy that micro-pores, remnants of the PM process, were observed in the Mo alloy BM. These gas-containing micro-pores may serve as potential sources for post-weld porosity defects.



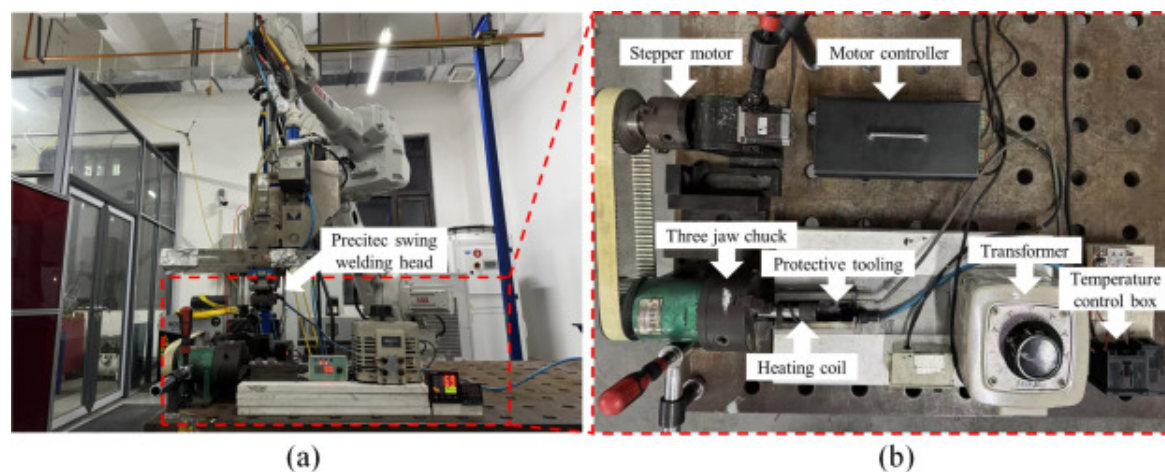
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Fig. 1. (a) Schematic diagram of the test material dimensions; (b) The microstructure of the axial direction of the pipe; (c) The microstructure of the radial direction of the pipe.

2.2. Experimental and modeling methods

The LW experimental system is illustrated in Fig. 2. The setup primarily consisted of an IPG YLS-8000 fiber laser, an ABB robot, a temperature-controlled heating and insulation device, a three-jaw chuck rotation fixture, and a welding protection device. The Precitec laser processing head featured a maximum output power of 4kW, a focal spot diameter of 400 μ m, a wavelength of 1070nm, and a beam focal length of 295mm.



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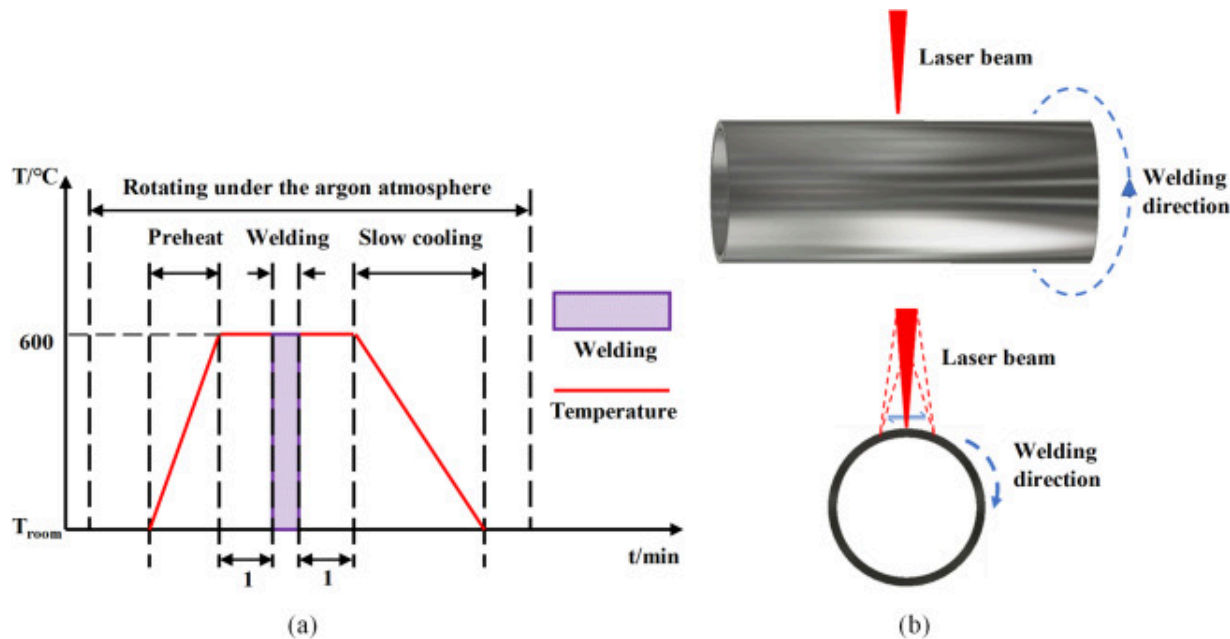
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Fig. 2. LW device. (a) Front view; (b) Top view.

The welding procedure was conducted as follows: (1) Before welding, the Mo alloy pipes were sequentially ground with 280# and 400# abrasive paper to remove the oxide layer, followed by ultrasonic cleaning in acetone and drying. (2) A pipe was secured using a three-jaw chuck rotating fixture, placed inside a protective chamber, and positioned by adjusting the laser defocus distance and spot location before closing the chamber. (3) Argon gas with a purity of 99.99% was introduced into the chamber at a flow rate of 30L/min for approximately 1.5min. (4) The pipe was preheated to 600°C and held at this temperature for 1 min [34]. (5) The rotating fixture and welding program were simultaneously initiated to commence the welding process. (6) Upon completion of

welding, the rotation was stopped, and the sample was maintained at 600°C for an additional 1 min. (7) Finally, the sample was cooled. Once the temperature dropped below 100°C, the argon gas supply was shut off, the chamber was opened, and the welded sample was retrieved, concluding the process.

In laser beam oscillation welding, common oscillation patterns include circular (“○”), figure-eight (“8”), infinity-shaped (“∞”), and linear back-and-forth motions. As mentioned above, remelting is beneficial for eliminating porosity in Mo alloy joints [10], and beam oscillation can be regarded as localized remelting over a small region. In addition, due to the equipment limitations, the investigation focuses specifically on the linear back-and-forth oscillation mode along the weld direction and its influence on porosity defects in Mo alloy welds. The welding procedure is illustrated in Fig. 3(a), and the beam oscillation pattern is shown in Fig. 3(b).



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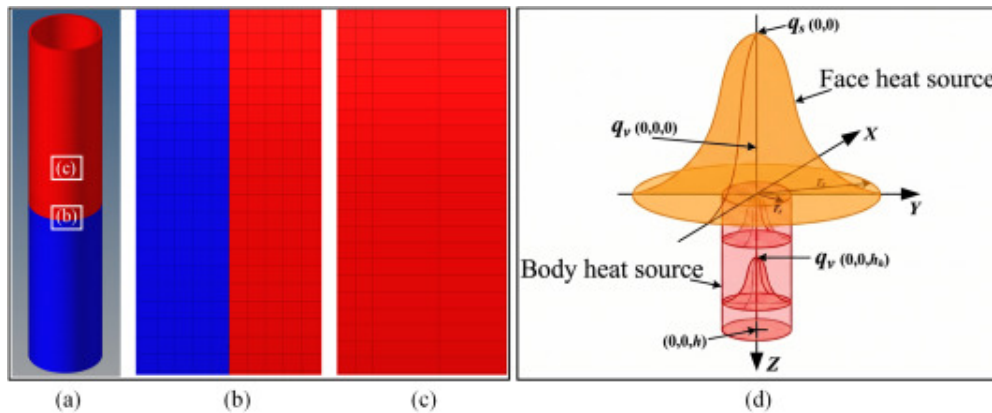
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Fig. 3. Schematic diagram of the welding procedure and the beam oscillation welding process. (a) Process flowchart; (b) Welding process diagram.

The geometric and mesh models of the Mo alloy heat pipe were established, as shown in Fig. 4(a)–(c). To balance calculation accuracy and computational efficiency, refined meshes with a size of 0.25mm were generated within the welding zone, while coarser meshes of 1.00mm were adopted in regions far from the weld bead. The temperature field was calculated. A composite heat source model combining a Gaussian surface heat source and a Gaussian cylindrical heat source [35] was employed to describe the laser heat source, as illustrated in Fig. 4(d). The heat distribution formulas of the surface and cylindrical Gaussian heat sources are defined in Eq. 1 and Eq. 2, respectively:

$$q_s(x, y) = \frac{\alpha Q_s}{\pi r_s^2} \exp \left[-\frac{\alpha(x^2 + y^2)}{r_s^2} \right] \quad (1)$$

$$q_v(x, y, z) = \frac{6Q_v(H - \beta z)}{\pi r_v^2 H^2 (2 - \beta)} \exp \left[-\frac{3(x^2 + y^2)}{r_v^2} \right] \quad (2)$$



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Fig. 4. Establishment of the finite element model for Mo alloy heat pipe welding: (a)–(c) 3D mesh model; (d) Heat source model.

The relationship between Q_s and Q_v is expressed by Eq. 3:

$$Q\eta = Q_s + Q_v \quad (3)$$

where α is the heat flux concentration coefficient, β denotes the attenuation coefficient, Q_s is the power of the surface heat source, Q_v is the power of the volumetric heat source, r_s represents the effective action radius of the surface heat source, r_v represents the effective action radius of the volumetric heat source, H is the effective penetration depth of the volumetric heat source, and η is the effective heat absorption coefficient of the material.

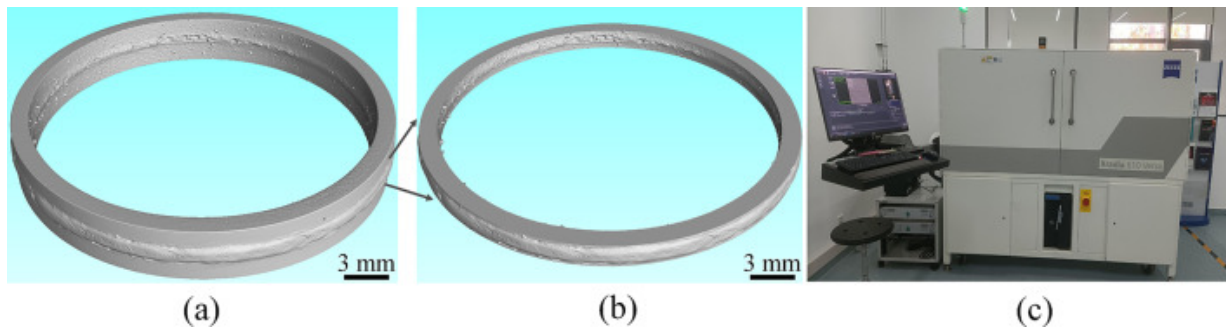
In this study, the temperature dependence of thermophysical parameters is neglected. The material property parameters [28] adopted in the finite element numerical simulation are listed in Table 1.

Table 1. Thermophysical property parameters.

Material Property	Value
Density (ρ)·g / cm ³	10.2
Melting point (T_m)·K	2895
Boiling point (T_b)·K	5077
Solid thermal conductivity (k_s)·W / (m·K)	138
Liquid thermal conductivity (k_l)·W / (m·K)	88
Specific heat capacity (c)·J / (K·kg)	250.78
Poisson's ratio	0.293

2.3. Analysis and test methods

The Computerized tomography (CT) specimens were prepared by cutting 1.5 mm from the weld center in both directions, resulting in a 3-mm-long segment centered on the weld bead, as shown in Fig. 5(a). The region selected for statistical CT analysis is indicated in Fig. 5(b). Pore defects in the weld bead were characterized using the Xradia 610 Versa X-ray imaging system illustrated in Fig. 5(c). CT data post-processing was performed using Dragonfly software.



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Fig. 5. (a) Schematic diagram of the CT specimen sampling; (b) Analysis zone; (c) Testing equipment.

After CT analysis, cross-sectional metallographic specimens of the joints were prepared following standard metallographic procedures. A mixed etchant consisting of 15 wt% $K_3[Fe(CN)_6]$ and 15 wt% NaOH solutions in a volume ratio of 2:1 was used for etching. The specimens were etched for 10 s, then thoroughly cleaned and dried. Microstructural observations of the joint cross-sections were conducted using a Nikon Eclipse MA200 optical microscope. A Hitachi SU8230 cold-field emission scanning electron microscope (SEM), equipped with an energy-dispersive X-ray spectrometer (EDS), was employed to examine the pore's morphology, and elemental distribution. Additionally, molten pool dynamics during LW were captured using the Revealer V2.0 high-speed imaging system at an acquisition rate of 4000 frames per second, combined with a dedicated backlight illumination source.

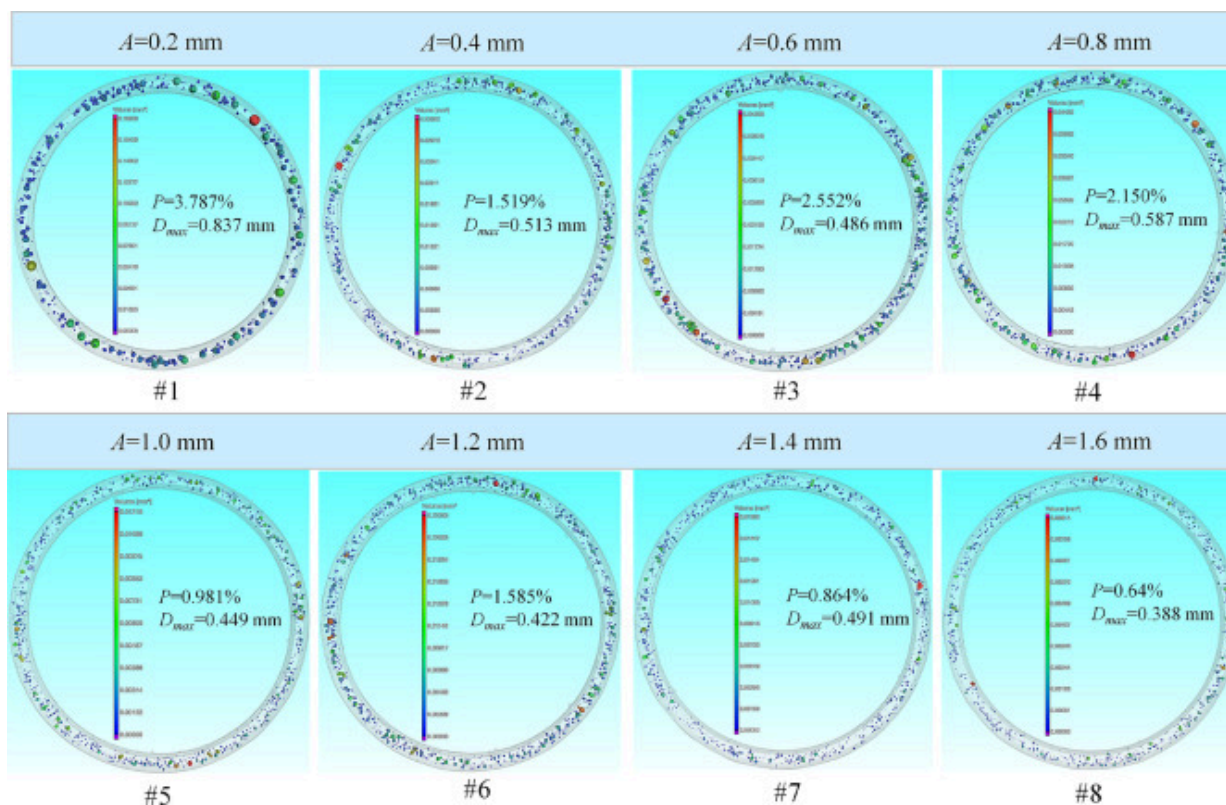
3. Results and discussion

3.1. Effect of beam oscillation amplitude on LWed pores of Mo pipes

To investigate the influence of beam oscillation amplitude on pore defects in welded Mo pipes, single-variable experiments were conducted according to the parameters in [Table 2](#), the welding rotational speed was fixed at 2r/s, and the welding duration was set to 2s. The distribution of pore defects is visualized in the contour maps in [Fig. 6](#). It can be visually observed that the pore defects are mitigated with increasing oscillation amplitude.

Table 2. Experimental parameters for the influence of beam oscillation amplitude on pore defects.

Num.	Laser Power P/W	Oscillation frequency f/Hz	Oscillation amplitude A/mm
#1	3200	500	0.2
#2		500	0.4
#3		500	0.6
#4		500	0.8
#5		500	1.0
#6		500	1.2
#7		500	1.4
#8		500	1.6



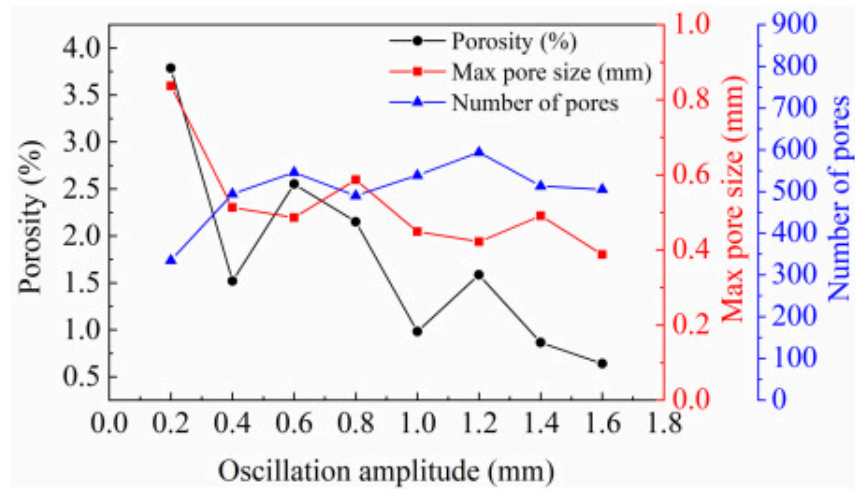
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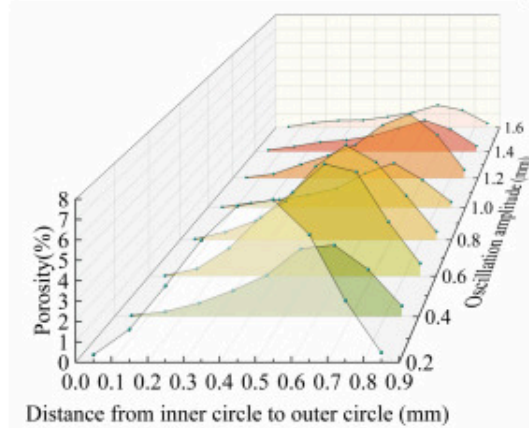
Fig. 6. Pore distribution under different beam oscillation amplitude. (P : porosity, D_{max} : Maximum equivalent circle diameter of pores).

Specifically, the porosity was calculated as the volume ratio of the total pores to the weld bead. It was then evaluated for different oscillation amplitudes at a fixed frequency. Fig. 7(a) presents the curve illustrating the influence of the beam oscillation amplitude on the pore defects. The statistical results show that, at a constant oscillation frequency of 500Hz, the weld joint porosity decreased from 3.787% to 0.640% as the oscillation amplitude increased from 0.2mm to 1.6mm. The overall trend indicates that the porosity of the Mo pipe weld joint gradually decreases with increasing beam oscillation

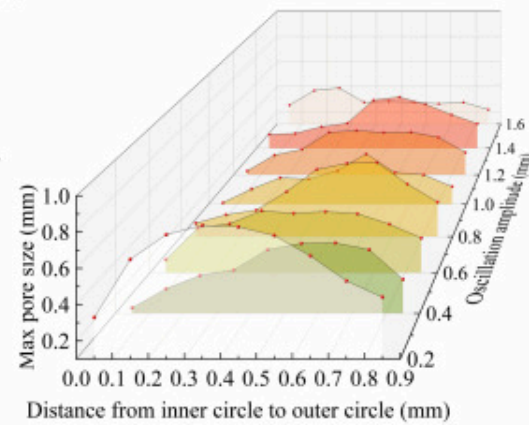
amplitude. Meanwhile, the maximum pore size correspondingly decreased with the increase in oscillation amplitude, dropping from 0.837mm to 0.388mm. In addition, the total number of pores in the weld joints was also counted. Except for the 0.2mm amplitude, which showed a significantly lower pore count, the number of pores fluctuated slightly between 500 and 600 under the other tested amplitudes.



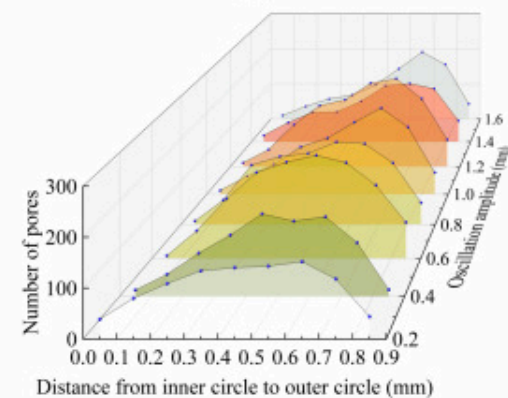
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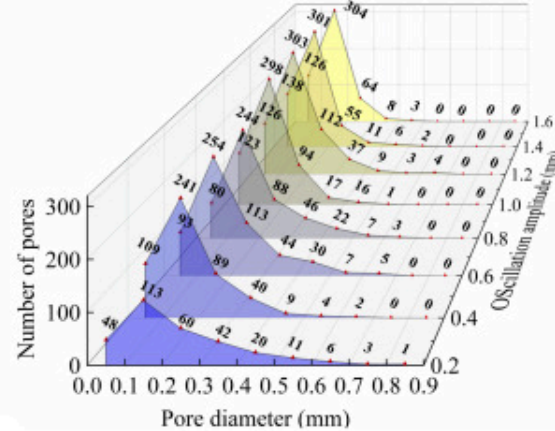
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(c)



(d)



(e)

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Fig. 7. Effects of beam oscillation amplitude on weld porosity, maximum pore size, and pore number. (a) Overall porosity, maximum pore size, and pore number; (b) Regional porosity distribution under different amplitudes; (c) Regional distribution of maximum pore size under different amplitudes; (d) Regional pore number distribution under different amplitudes; (e) Number of pores by size under different oscillation amplitudes.

Furthermore, these Mo alloy weld joints were divided into 9 regions from the inner wall to the outer wall at 0.1 mm intervals to further investigate the spatial distribution difference of pore characteristics under different oscillation parameters. The porosity, maximum pore size, and pore number in each region were statistically analyzed. As shown in [Fig. 7\(b\)-7\(d\)](#), for all oscillation amplitudes, the porosity, maximum pore size, and pore number along the path from the inner to the outer wall exhibited a consistent trend of first increasing and then decreasing. The peaks of porosity and pore number were predominantly distributed in the region of 0.5–0.7 mm from the inner wall towards the outer wall, while the spatial distribution of the maximum pore size showed no significant regularity.

Additionally, a clear inverse correlation was observed between pore size and number for all pores exceeding 0.072 mm, i.e., 8% of the wall thickness. Specifically, [Fig. 7\(e\)](#) shows that smaller pores were significantly more numerous than larger ones under all tested oscillation amplitudes.

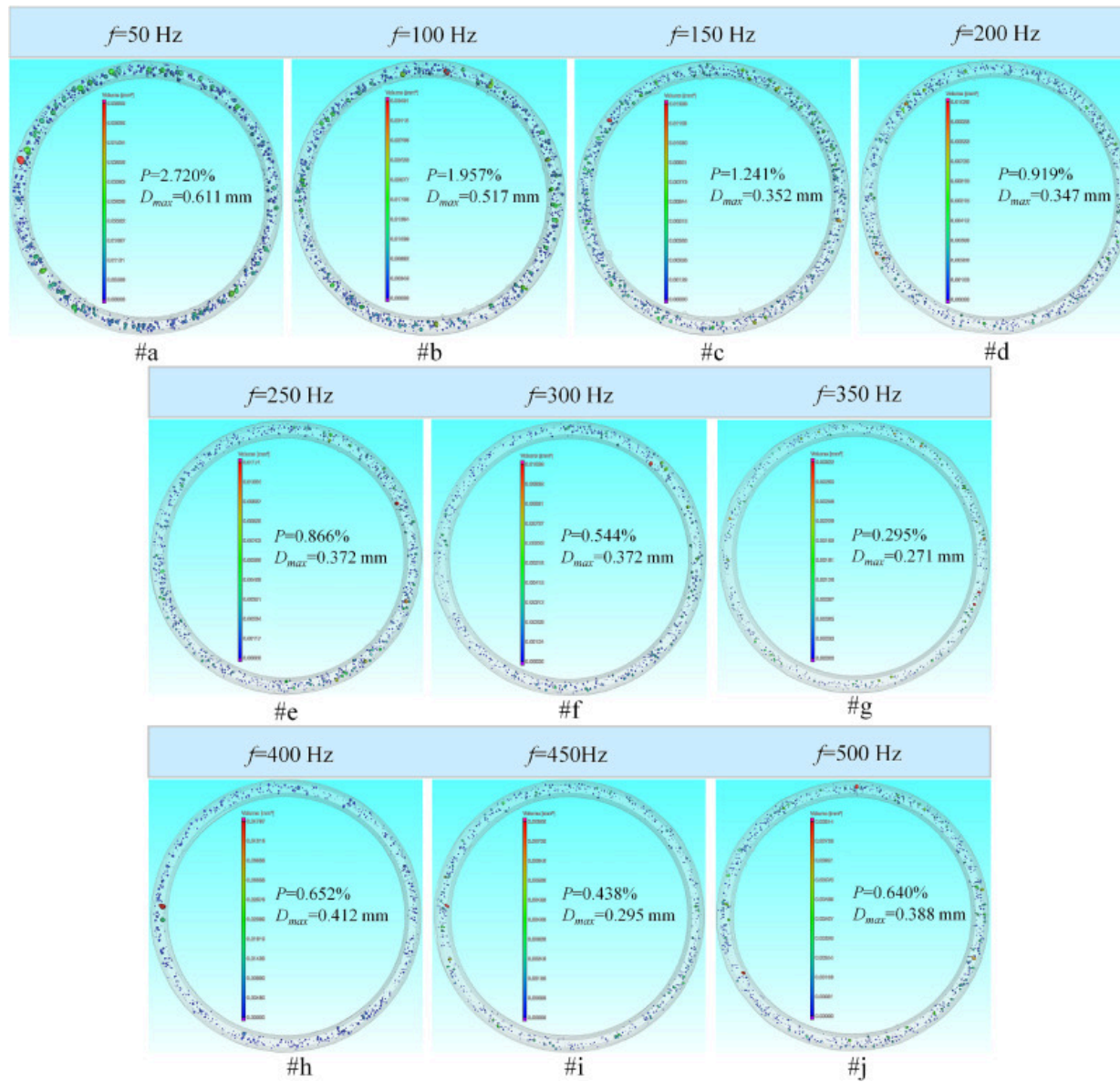
3.2. Effect of beam oscillation frequency on LWed pores of Mo pipes

Concurrently with the amplitude study, the effect of beam oscillation frequency on pore defects was investigated according to the welding parameters provided in [Table 3](#).

Table 3. Experimental parameters for the influence of beam oscillation frequency on pore defects.

Num.	Laser Power P/W	Oscillation frequency f/Hz	Oscillation amplitude A/mm
#a	3200	50	1.6
#b		100	1.6
#c		150	1.6
#d		200	1.6
#e		250	1.6
#f		300	1.6
#g		350	1.6
#h		400	1.6
#i		450	1.6
#j		500	1.6

Fig. 8 presents the contour maps of pore defect distribution in Mo pipe weld joints under different oscillation frequencies. It can be observed that as the beam oscillation frequency increases, the number of pore defects gradually decreases, with a particularly noticeable reduction in large-sized pores. It is visually evident that the improvement in pore defects is most significant when the oscillation frequency reaches 300–450Hz.

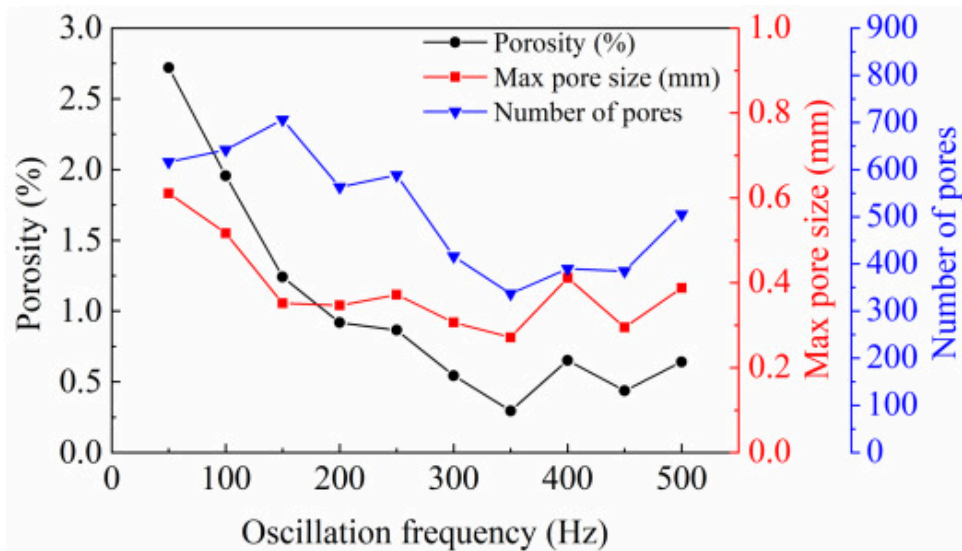


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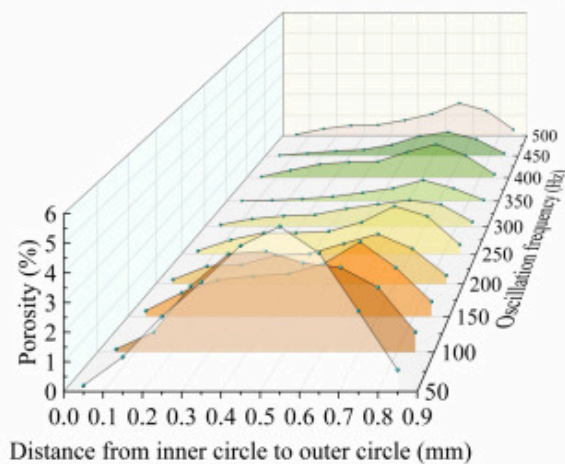
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Fig. 8. Pore distribution under different beam oscillation frequencies.

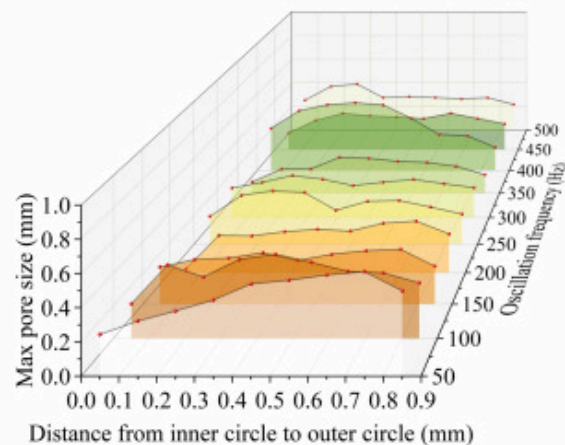
Fig. 9(a) presents the curves illustrating the influence of beam oscillation frequency on the overall porosity, maximum pore size, and pore number in the weld joints. The results indicate that the porosity, maximum pore size, and pore number exhibit a consistent trend, initially decreasing and then slowly increasing. The minimum values were achieved at an oscillation frequency of 350Hz, with a porosity of 0.295%, a maximum pore size of 0.271 mm, and a pore count of 337.



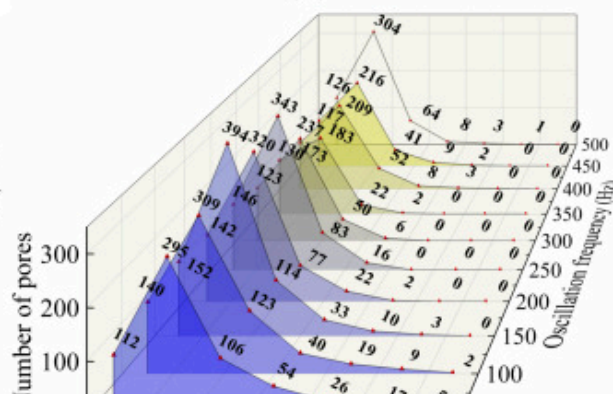
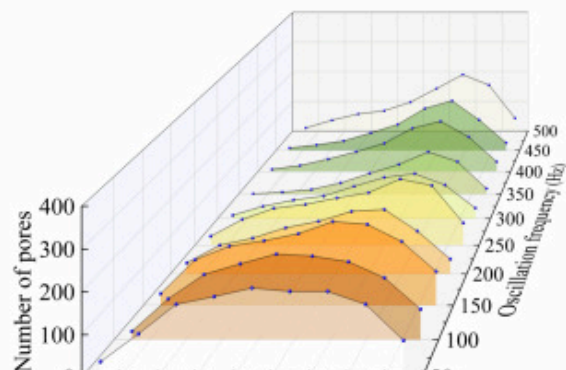
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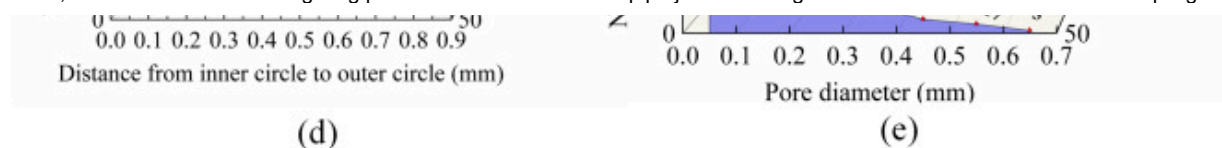


(b)



(c)





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Fig. 9. Effects of beam oscillation frequency on weld porosity, maximum pore size, and pore number. (a) Overall porosity, maximum pore size, and pore number; (b) Regional porosity distribution under different frequencies; (c) Regional distribution of maximum pore size under different frequencies; (d) Regional pore number distribution under different frequencies; (e) Number of pores by size under different oscillation frequencies.

Similarly, the spatial distributions of porosity, maximum pore size, and pore number from the inner to the outer wall were analyzed. As shown in Fig. 9(b)–(d), the distribution patterns are consistent with those observed under different amplitudes at a fixed frequency, exhibiting an initial increase followed by a decrease from the inner to the outer wall, with peaks concentrated in the region of 0.5–0.7 mm from the inner wall. However, the distribution of the maximum pore size again showed no clear regularity and appeared relatively random.

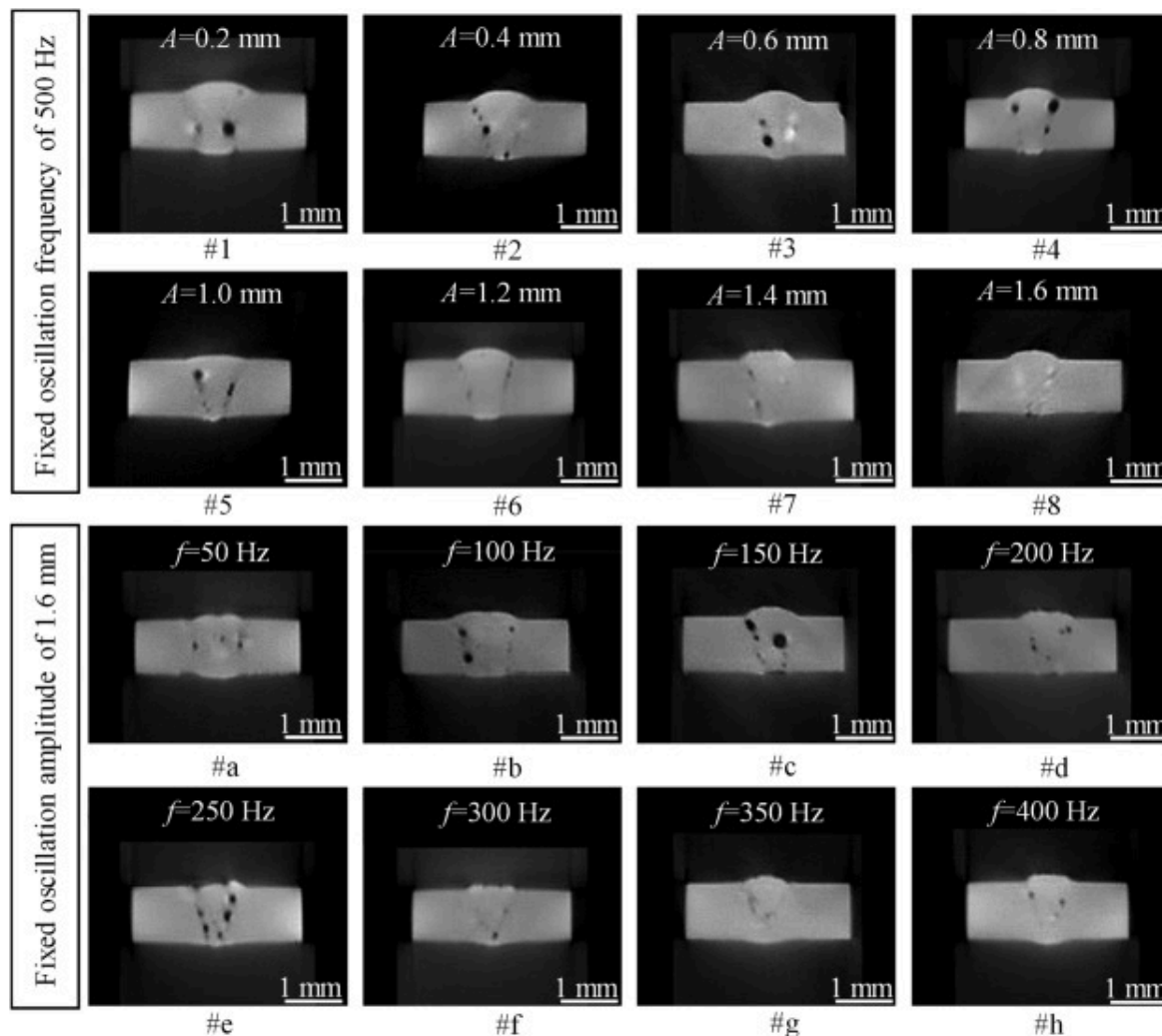
3.3. Analysis of the distribution characteristics of pores

Statistical analysis of the porosity, and pore number distribution revealed a unimodal distribution along the weld deep direction (from the inner to the outer wall), which first increased and then decreased. The peak was consistently located in the mid-upper region, 0.5–0.7 mm from the inner wall. This spatial distribution pattern was not significantly influenced by the laser beam oscillation amplitude or frequency.

In fact, the bottom of the laser weld pool is the narrowest, with relatively high keyhole stability and a small molten volume. This results in a low absolute probability for pore nucleation. Furthermore, the buoyancy force is greater at the bottom than in the mid-

upper region, causing any formed pores to rise rapidly. Conversely, pores generated in the upper region have the shortest vertical distance to the free surface and can escape almost instantaneously. However, in the mid-upper region, pores that have overflowed upwards from lower regions experience a prolonged upward path, and the buoyancy gradually decreases as it approaches to the top of the molten pool. They are thus highly susceptible to being captured by the rapidly advancing lateral solidification front. Consequently, this leads to the peak values in porosity and pore number observed in this specific zone.

Furthermore, the pore distribution in the weld cross-sections under various oscillation parameters was obtained, as shown in [Fig. 10](#). From the figure, it can be observed that, regardless of the welding parameters, a significant number of pores are consistently distributed near the fusion line.



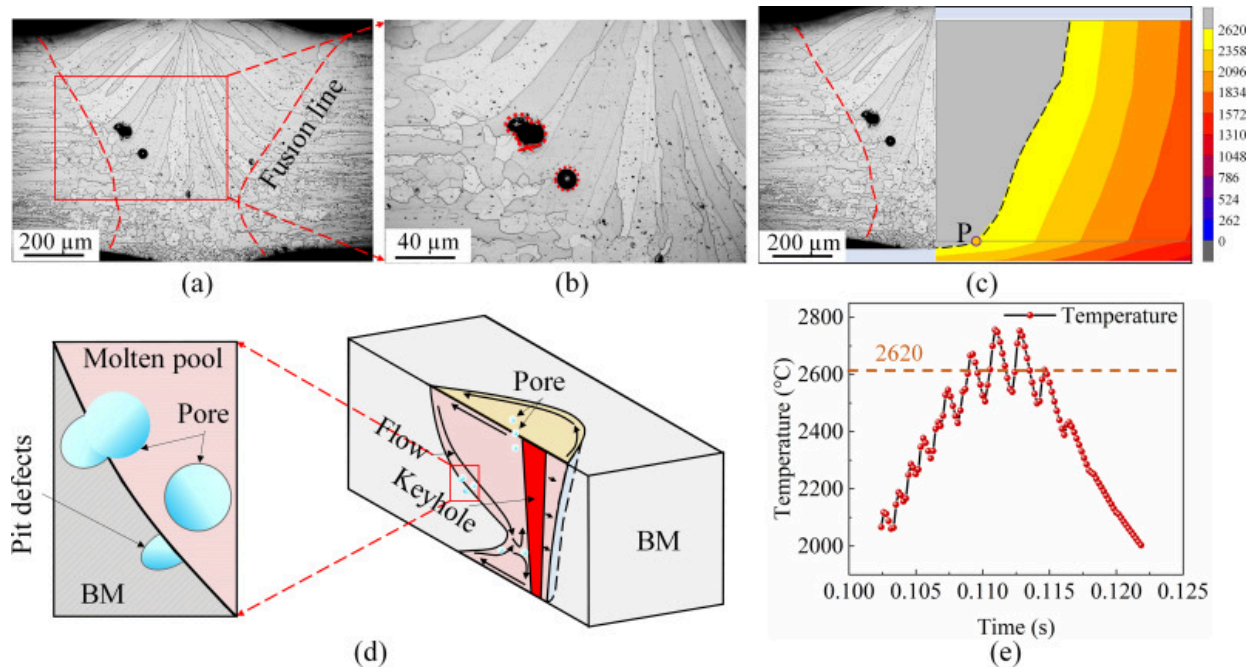
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Fig. 10. Cross-sectional pore distribution in weld joints under different laser beam oscillation parameters. (#1–#8 and #a–#h correspond to the oscillation parameters in Table 2 and Table 3, respectively).

Dong et al. [9] demonstrated that Mo alloys prepared by PM are easier to generate pores during welding, and with pores concentrated near the fusion line. Zhang et al. [10] proposed that welding pores in Mo alloys may differ from conventional metallurgical and process-related pores, instead being a base-metal-released type pores associated with the PM process. This type of welding pore has also been observed in the weld joints of other PM prepared materials [36]. For PM-Mo alloy, the BM inevitably contains micro-voids, which can readily serve as heterogeneous nucleation sites for pores [9].

As shown in Fig. 11(a) and (b), pores near the fusion line are clearly visible and are distributed along the grain boundaries (GBs). It has been proposed that during the welding of Mo alloys manufactured by PM, pit defects exist at the molten pool boundary [7], as shown in Fig. 11(d). These so-called pit defects are essentially gas-filled micropores in the BM. The micropore wall on the molten pool side melts under welding heat, while the high-pressure gas trapped inside is heated and expands outward to form a bulge. Meanwhile, the metal matrix on the BM side remains in an unmelted state, resulting in the typical gourd-shaped pore morphology. The heterogeneous nucleation of pores is concentrated mainly on these pit defects at the pool edge, Fig. 10(d) schematically illustrates this process. Besides, due to the highest temperature at the pool center and a relatively lower temperature near the fusion line, a surface tension gradient is established (higher temperature corresponds to lower surface tension). This gradient induces Marangoni convection, whereby generated pores experience a Marangoni force pushing them from the hot pool center towards the cooler fusion line. This is consistent with the conclusions of other researchers [37], [38].



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Fig. 11. (a-b) Distribution of pore defects in the cross-section of the weld bead; (c) Temperature field nephogram of weld cross-section; (d) Schematic diagram of the pore formation process; (e) Time-temperature curve at point P with temperature above 2000°C.

Further, the molten pool near the fusion line exhibits relatively high viscosity and poor fluidity. The solidification front initiates from the fusion line and advances rapidly, a phenomenon exacerbated by the high melting point of Mo alloys and the inherent high solidification rate in LW. This rapidly advancing solidification front quickly overtakes the pore interface, trapping the pores and consequently leading to the easy formation of pores adjacent to the fusion line.

Additionally, due to the significant difference in solubility of gaseous elements between solid and liquid Mo, the BM in the heat-affected zone (HAZ), though unmelted, is heated to an extremely high temperature. Fig. 11(c) presents the temperature field nephogram of the weld cross-section during welding. It can be observed that the temperature range of the HAZ is 2358°C–2620°C, equivalent to 0.9–1.0 T_m of Mo. Under such high-temperature conditions, the diffusion rates of gaseous atoms such as oxygen (O), hydrogen (H), and nitrogen (N) increase significantly, which provides sufficient kinetic conditions for atomic migration.

According to the Arrhenius Eq. [39] and the simplified formula for diffusion distance, as shown in Eq. 4 and Eq. 5:

$$D = D_0 \exp\left(-\frac{Q}{RT}\right) \quad (4)$$

$$x = 2\sqrt{Dt} \quad (5)$$

where D is the diffusion coefficient, D_0 is the pre-exponential factor, Q is the activation energy for diffusion, R is the gas constant, T is the temperature, x is the root-mean-square diffusion distance, and t is the high-temperature dwell time.

Assuming constant physical parameters independent of temperature, using the median values of $D_0=0.05\text{cm}^2/\text{s}$ and $Q=26\text{kcal/mol}$ from reference [40] to roughly estimate the diffusion behavior of O atoms during welding. The diffusion coefficient at 2358°C is approximately calculated to be $3.46\times 10^{-8}\text{m}^2/\text{s}$.

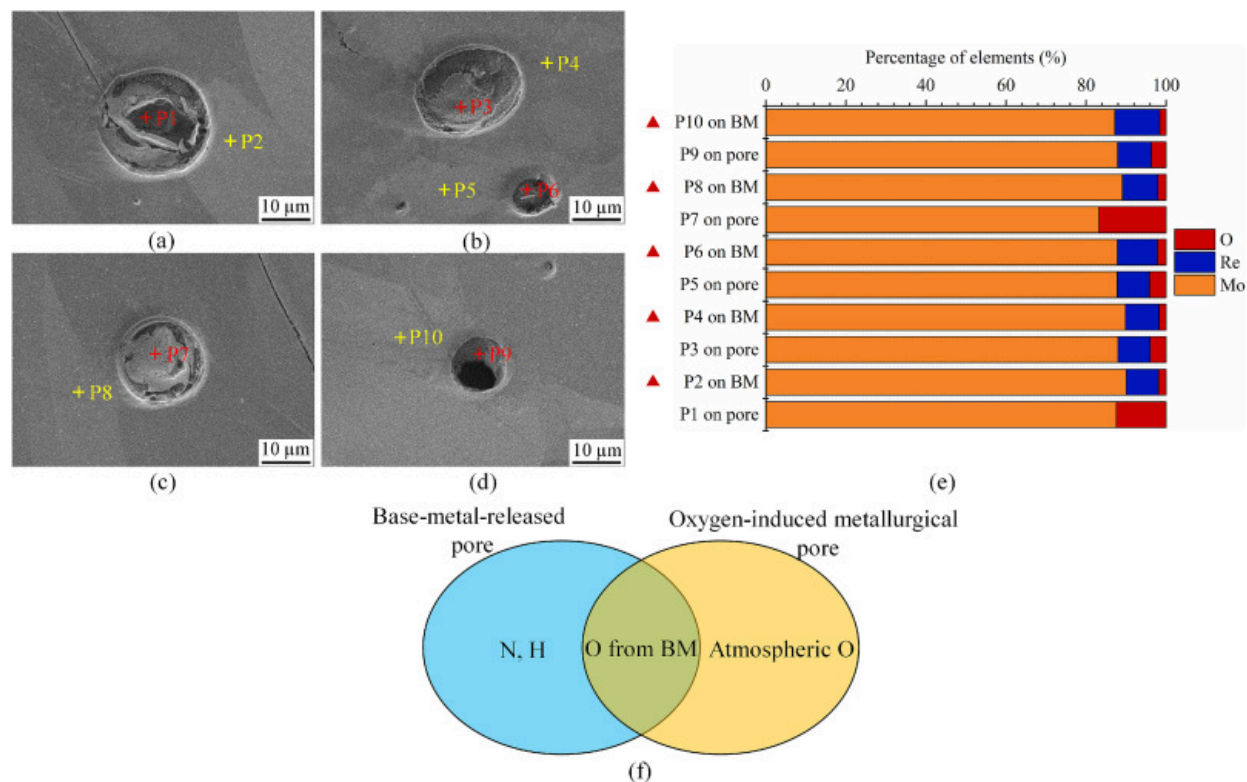
Since the cooling rate of the weld bottom surface is the fastest, the time-temperature curve within one welding cycle was extracted, as shown in Fig. 11(e). It can be seen from the curve that the duration of the point above the melting point during welding is 2.036ms. Since a total of 4 welding cycles are performed, the total duration above the melting point is 8.144ms, while the duration above the melting point in other regions is longer. Substituting the relevant parameters into Eq. (4), (5), the calculated diffusion distance is approximately 34μm, which is larger than some of the grain sizes of the HAZ.

Considering factors such as GBs and dislocations in the BM, the actual diffusion distance may be longer, which thermodynamically confirms the feasibility of gas atom diffusion from the BM to the molten pool.

Concurrently, owing to the high gas concentration within the Mo alloy BM, a strong concentration gradient and diffusion flux of these elements exist between the BM side and the molten pool side. Consequently, a powerful driving force is established for gas diffusion from the high-concentration BM side towards the molten pool. The gas tends to diffuse along pathways such as GBs and is released at the fusion line interface. This leads to the supersaturation of gaseous elements in the liquid metal near the fusion line. Owing to the significant difference in gas solubility between solid and liquid molybdenum alloys, the supersaturated liquid metal near the fusion line tends to precipitate dissolved gas rapidly during solidification. Such gas preferentially nucleates and grows at pre-existing micropores, GBs, and other heterogeneous nucleation sites, eventually forming pore defects and distributing along the fusion line.

Further EDS analysis of the pores near the fusion line was conducted, and the results are shown in Fig. 12(a-e). It can be seen that the oxygen (O) content on the inner surfaces of the pores is higher than that in the surrounding BM. Specifically, the O concentrations at points P1 and P7 are significantly elevated compared to other locations, and no rhenium (Re) was detected at these sites, strongly suggesting the formation of Mo oxides. This indicates that the pores are oxygen-induced. In fact, distinguishing between base-metal-released pores and oxygen-induced metallurgical pores relies not only on EDS results but also on pore distribution characteristics. Fig. 12(f) illustrates the classification of pore types. Pores concentrated near the fusion line are predominantly base-metal-released pores; among them, those with high oxygen content simultaneously belong to oxygen-induced metallurgical pores. In contrast, isolated pores with high oxygen content far from the fusion line are defined as pure oxygen-induced metallurgical pores. The results indicate that purely oxygen-induced metallurgical pores can be effectively eliminated by beam oscillation, and the remaining defects are mainly base-metal-released pores distributed near the fusion line. Note that in Fig. 11(a-b), these pores exhibit a distinct

tendency to distribute along GBs. As reported [41] that O atoms energetically prefer to segregate at sites in both the GB core and the vicinity of the GB plane. Consequently, during welding, grain boundaries enriched with segregated oxygen are prone to pore formation.



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Fig. 12. (a-d) EDS scanning position; (e) EDS scan results; (f) Classification of base-metal-released pores and oxygen-induced metallurgical pores.

Meanwhile, both Fig. 7(e) and Fig. 9(e) show a trend where smaller pore sizes correspond to a greater number of pores. This observation is also related to the base-metal-released pore characteristic of PM-Mo alloys during welding. The residual gaseous elements are uniformly dispersed within the BM, creating numerous, discrete pore nucleation sites

throughout the molten pool. According to Stokes' Law [42], as shown in Eq. 6:

$$V_f = \frac{2(\rho_1 - \rho_2)gr^2}{9\mu} \quad (6)$$

Where r is the bubble radius, ρ_1 is the density of the molten metal, ρ_2 is the density of the gas inside the pores, g is the acceleration due to gravity, μ is the dynamic viscosity of the molten metal.

For the same material and welding process, the pore rise velocity is proportional to the square of the pore radius. Therefore, the overflow velocity of micron-sized pores is extremely slow, making them nearly immobile before the molten pool solidifies. Simultaneously, due to the large distances between them and the short time available, adjacent small pores lack sufficient time to coalesce into larger ones. Consequently, upon rapid solidification, a large number of these small pores, which were nucleated but had no time to grow or overflow, remain trapped in the weld joints.

3.4. The mechanism of longitudinal beam oscillation on pore mitigation in LW of Mo alloy pipes

Although beam oscillation cannot change the spatial distribution pattern of pores or the size distribution trend in Mo alloy LW, it still effectively improves the two most critical metrics for structural components of Mo alloy on pore defect: the overall porosity and the maximum pore size.

At a fixed oscillation frequency and under the condition that the weld depth is larger than the wall thickness of the pipe, an increase in the oscillation amplitude elongates the laser-scanned zone. Considering a specific point P on the weld bead, its total residence time within the laser scanning zone under stable beam oscillation welding is given by Eq. 7:

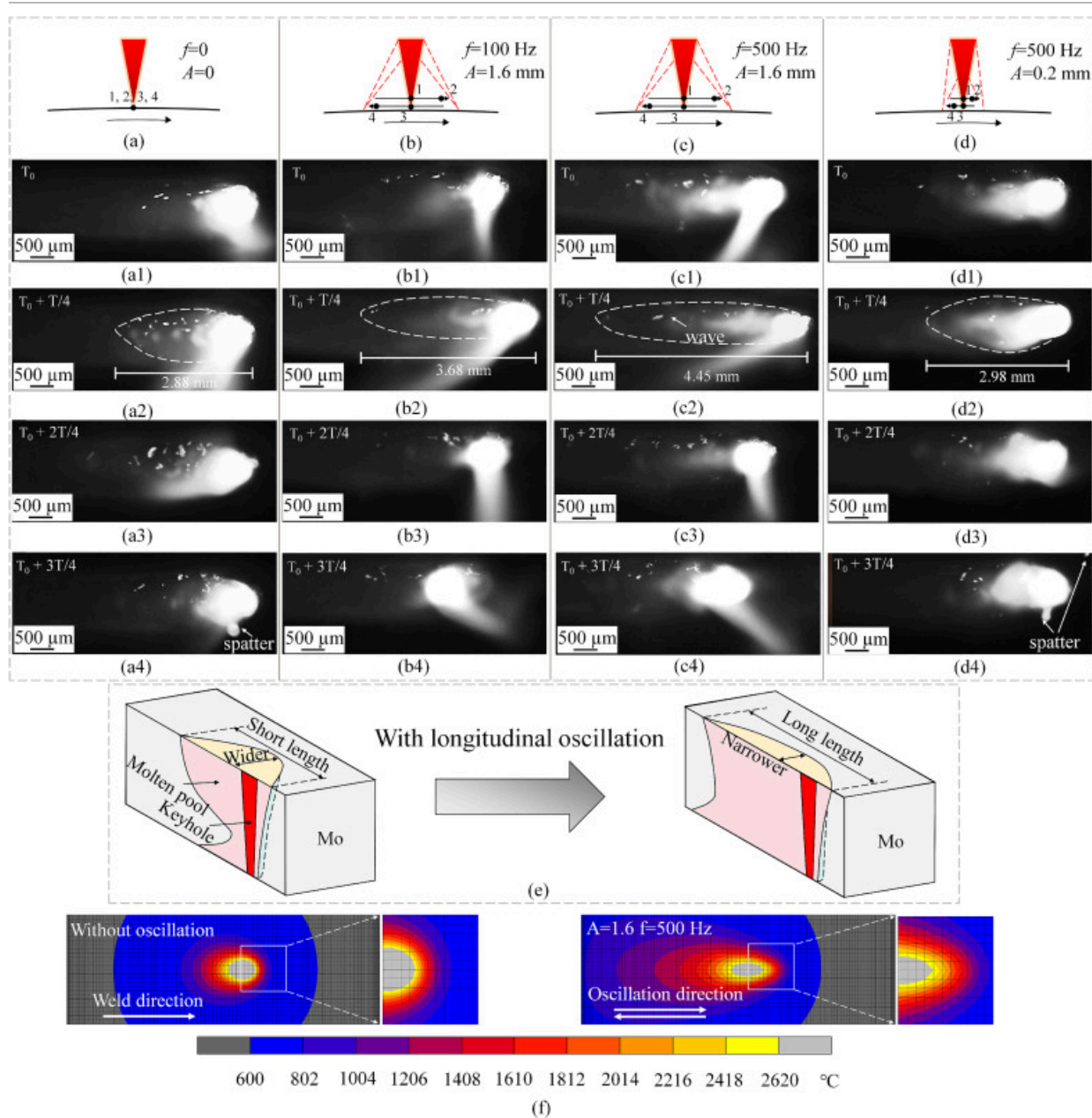
$$\Delta t = \frac{A}{V} \quad (7)$$

Where Δt is the total residence time of point P on the pipe under the laser beam

oscillation scanning zone, A is the oscillation amplitude, V is the rotational speed of the pipe.

It can be concluded that, with the rotational speed of the pipe held constant, the residence time of any point on the pipe within the laser scanning zone is proportional to the oscillation amplitude. This means that an increase in amplitude extends the time that any point on the weld bead remains within the scanning zone, thereby providing a longer time for pores to overflow.

A comparison of the molten pool morphology captured by high-speed photography within one oscillation cycle under different parameters is shown in Fig. 13. Without oscillation in Fig. 13(a), the pool was wider and shorter, with a length of approximately 2.88mm, accompanied by spatter generation. When the amplitude was 1.6mm at a frequency of 100Hz, the pool became narrower and longer, with a length of about 3.68mm in Fig. 13(b). When the frequency was increased to 500Hz, the pool length further increased to 4.45mm in Fig. 13(c). However, in Fig. 13(d) when the frequency was maintained at 500Hz and the amplitude was reduced to 0.2mm, the pool length decreased to 2.98mm, nearly identical to the case without oscillation, but spatter still occurred under this small-amplitude high-frequency condition.



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Fig. 13. High-speed photographic morphology of the molten pool within one beam oscillation cycle under different oscillation parameters. (a) $A=0, f=0$; (b) $A=1.6, f=100$; (c) $A=1.6, f=500$; (d) $A=0.5, f=500$; (e) Schematic diagram of the molten pool evolution after introducing longitudinal oscillation; (f) Comparison of welding temperature fields with and without beam oscillation.

As shown in Fig. 13(e), after introducing longitudinal oscillation, the molten pool length increases while its width decreases. Comparative analysis was performed between without oscillation and with an oscillation amplitude of 1.6mm at a frequency of 500Hz. Fig. 13(f) shows the temperature field nephogram on the top surface of the weld, the gray areas represent regions where the temperature exceeds the melting point of Mo, which is 2620°C. The findings indicate that oscillation not only elongates the molten zone but also smooths the temperature gradient, particularly in the front of the welding pool advancing direction. This improves the stability of the molten pool, further reducing process-induced pores [43] and preventing cracking in Mo alloy weld bead.

The high-speed photography results under high-frequency oscillation also confirm that an increase in amplitude prolongs the molten pool length. This extended length can increase the overflow time for pores in the molten pool. In fact, longitudinal oscillation can be regarded as a form of remelting welding, which has been proven to be effective in reducing pore defects [44], [45].

Furthermore, beyond prolonging the pore existence time to facilitate their overflow, it has been reported that the moving keyhole can absorb pores and serve as an overflow channel for them [46]. Therefore, the influence on pores can also be analyzed by examining the number of times that a specific point P on the weld bead is scanned by the keyhole.

During the time interval Δt , the laser beam is subjected to a continuous oscillation. The oscillation frequency of the laser is f , meaning it completes f full back-and-forth cycles per second. Therefore, the number of scanning completed within the time Δt is given by Eq.

8:

$$N = 2f \Delta t = \frac{2fA}{V} \quad (8)$$

Where N is the number of laser scans, f is the oscillation frequency, A is the oscillation amplitude, V is the pipe rotational speed, Δt is the residence time of a point on the weld bead within the scanning zone.

It can be concluded that, with the workpiece rotational speed and oscillation amplitude held constant, the number of laser scans is also proportional to the oscillation frequency. This implies that increasing the oscillation frequency simultaneously increases the number of times a point P on the weld bead is remelted by the laser, i.e., the number of times it is scanned by the keyhole. This means that any generated pores have a greater probability of interacting with the keyhole, being absorbed or overflowing through it.

Since changing the oscillation amplitude and frequency within a certain range can both yield positive effects, the significance of their influence on porosity and pore size was further investigated. To eliminate the influence of dimensionality, the different amplitudes, frequencies, and their corresponding porosity and maximum pore sizes were normalized. The normalization formula is as Eq. 9 shows:

$$X_{norm} = (X - X_{min}) / (X_{max} - X_{min}) \quad (9)$$

Where X is the input variable (amplitude or frequency), X_{norm} is the normalized value, X_{min} and X_{max} are the minimum and maximum values of the variable, respectively.

A Bayesian method, which is more suitable for small-sample statistics, was employed in conjunction with Gaussian Process Regression to perform functional distribution probability statistics on the normalized data [47]. The Gaussian Process Regression formula is as Eq. 10 shows:

$$f(X) \sim GP(m(X), k(X, X')) \quad (10)$$

Where $f(X)$ is the unknown function distribution to be obtained, GP denotes a Gaussian Process, $m(X)$ is the mean function and assumed to be $m(X)=0$, $k(X, \mathbf{X}')$ is the kernel function, X and $\mathbf{X}'$ are two values of the input variable. This study adopts the radial basis function (RBF) kernel and Matern 3/2 kernel with different smoothness prior assumptions for fitting and comparative analysis, as shown in Eq. 11 and Eq. 12:

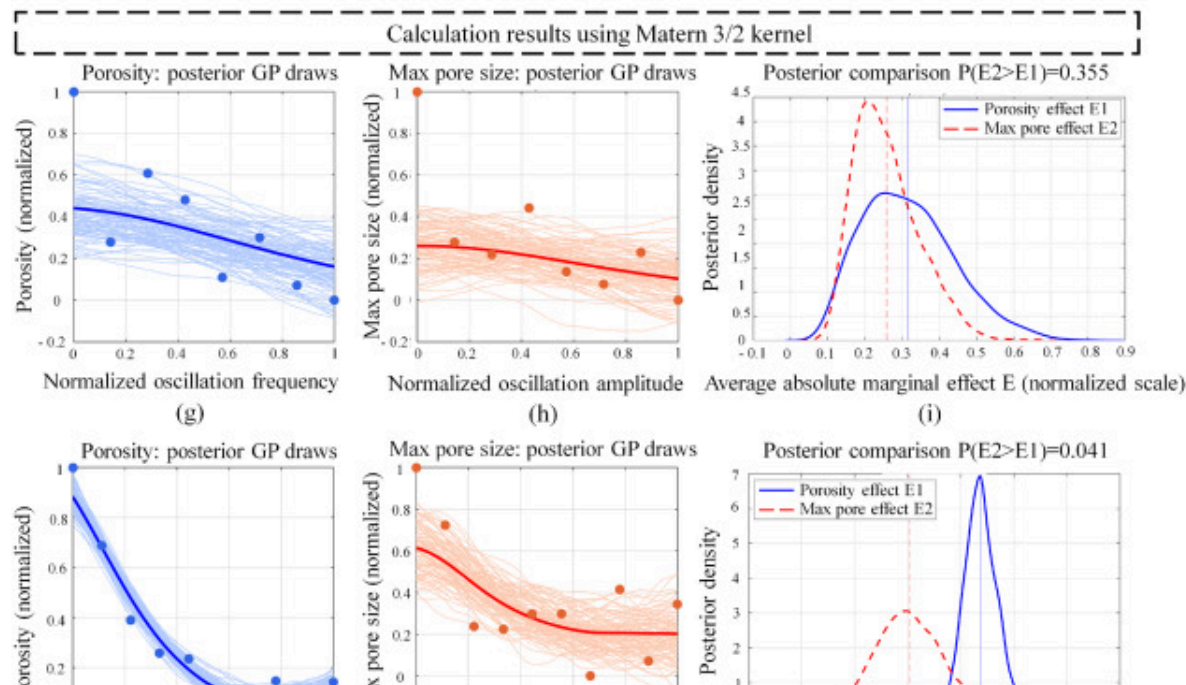
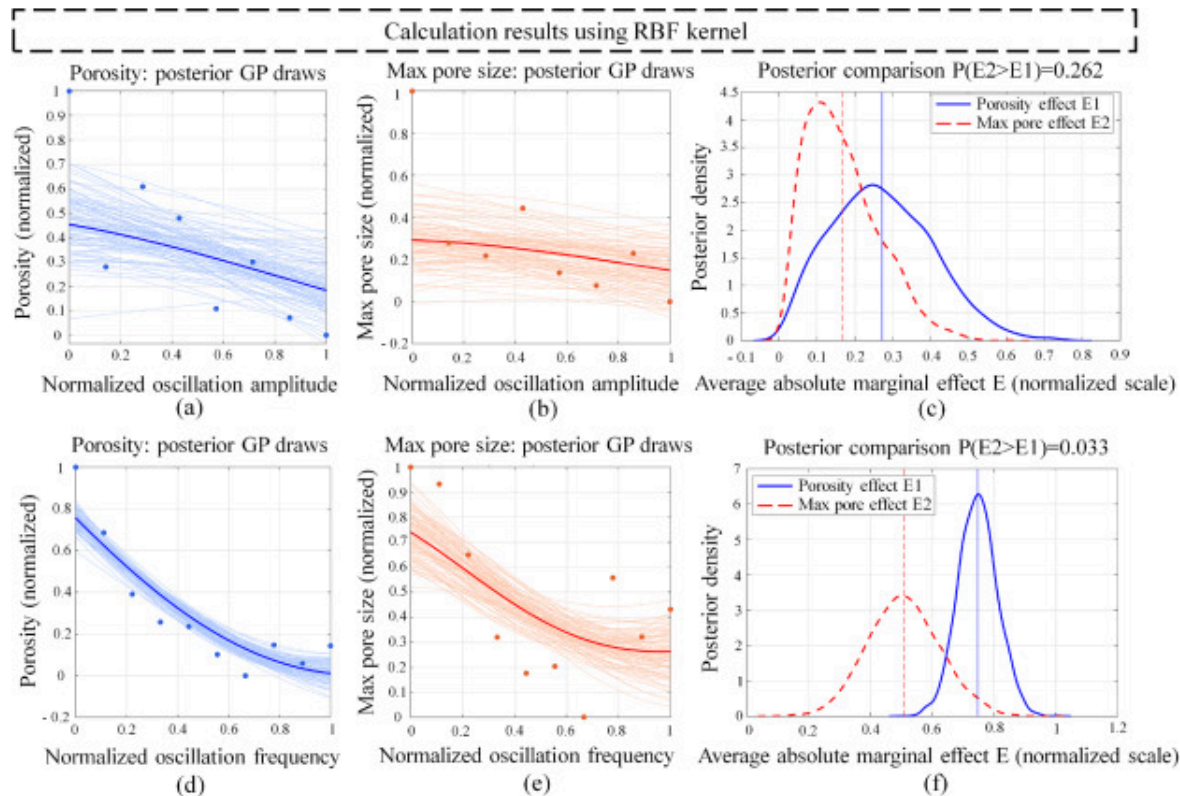
$$k(X, \mathbf{X}') = \sigma_f^2 \exp\left(-\frac{(X - \mathbf{X}')^2}{2l^2}\right) \quad (11)$$

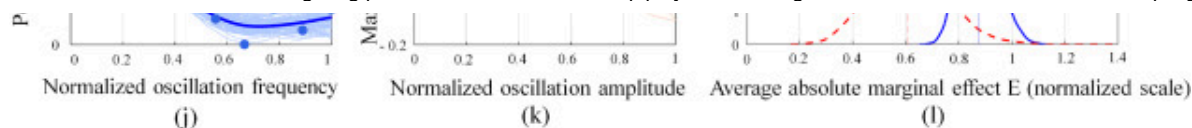
$$k(X, \mathbf{X}') = \sigma_f^2 \left(1 + \frac{\sqrt{3}}{l}d\right) \exp\left(-\frac{\sqrt{3}}{l}d\right) \quad (12)$$

Where σ_f denotes the signal standard deviation, controlling the overall variance of the output, and l represents the length scale, governing the sensitivity of the response to changes in the input, d is Euclidean distance between two points $\mathbf{X}$ and $\mathbf{X}'$. Both hyperparameters of the kernel function are optimized by maximizing the marginal likelihood, which balances the model's fitting capability with its generalization performance.

The detailed implementation is provided in the supplementary information. In addition, leave-one-out cross-validation (LOO-CV) was conducted with oscillation amplitude and oscillation frequency taken as input variables, respectively. The detailed results are presented in the supplementary information. LOO-CV results demonstrate that the samples causing model trend distortion are concentrated at both ends of the data range, which are typical high-leverage points. For small-sample datasets, fitting distortion after eliminating high-leverage points is a common problem of Gaussian Process Regression. Combined with diverse prior assumptions and LOO-CV verification, the good robustness of the proposed model is effectively validated.

The posterior sampling curves of the Gaussian process and the posterior distributions indicating effect significance are shown in Fig. 14.





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Fig. 14. Fitting curves and posterior distribution probabilities after normalization of the oscillation parameters on pore defects. RBF kernel: (a) Normalized porosity and (b) Maximum pore size versus normalized amplitude: Gaussian process posterior sampling curves; (d) Normalized porosity and (e) Maximum pore size versus normalized frequency: Gaussian process posterior sampling curves; (c) Comparison of marginal posterior distributions for the effect significance of amplitude and (f) Frequency on porosity and maximum pore size. Matern 3/2 kernel: (g-l) corresponds to (a-f).

Fig. 14(a), (b) and Fig. 14(g), (h) present the Gaussian process posterior sampling of the normalized data for porosity and maximum pore size, respectively, under different oscillation amplitudes at a fixed oscillation frequency. The probability densities of the mean gradients for the actual functional distributions in Fig. 14(a), (b) and Fig. 14(g), (h) are statistically analyzed and shown in Fig. 14(c), (i). Under Bayesian statistics, the marginal posterior distributions representing the effect significance of oscillation amplitude on porosity (Effect 1, E1) and maximum pore size (Effect 1, E2) exhibit large overlap, with their peak effect sizes being very close. The Bayesian posterior probability calculation indicates that the probability of the oscillation amplitude having a greater influence on the porosity than on the maximum pore size is 26.2% under the RBF kernel, and 35.5% under the Matern 3/2 kernel. This result implies that there is insufficient evidence ($P < 5\%$ or $P > 95\%$) to conclude a statistically significant difference in the effect of oscillation amplitude on these two responses. In other words, changing the oscillation amplitude is likely to influence both porosity and maximum pore size to a nearly identical degree.

Similarly, Fig. 14(d), (e) and Fig. 14(j), (k) display the Gaussian process posterior sampling of the normalized data for porosity and maximum pore size, respectively, under different oscillation frequencies at a fixed amplitude. The probability density distributions of the mean gradients are presented in Fig. 14(f), (l). Bayesian model inference reveals a distinctly different pattern in the effect of oscillation frequency.

The posterior distribution for the effect significance on porosity (E1) is situated almost within a region of higher probability density and shows a marked separation from that for the effect on the maximum pore size (E2). The posterior probability calculation yields $P(E2 > E1) = 0.033$ under the RBF kernel, and 0.041 under the Matern 3/2 kernel. This indicates that, with strong Bayesian confidence scores of 96.7% (RBF kernel) and 95.9% (Matern 3/2 kernel), increasing the oscillation frequency exerts a significantly stronger effect on reducing porosity than on decreasing the maximum pore size within the investigated parameter range. Therefore, oscillation frequency can be considered a highly effective process parameter for controlling porosity in Mo alloy LW process.

Furthermore, a direct comparison between the data in Fig. 14 (a), (b), (g), (h) (amplitude) and Fig. 14(d), (e), (j), (k) (frequency) intuitively reveals that the gradients associated with the normalized frequency are larger for both porosity and maximum pore size. This observation further indicated that oscillation frequency exerts a more visible influence on these responses than oscillation amplitude. It is also noteworthy that the gradient of the curves in Fig. 14(d), (j) and Fig. 14(e), (k) show signs of deceleration, suggesting that the effect of oscillation frequency may be approaching an inflection point.

By comparing the marginal posterior distributions of the significance of the effects of the process parameters on the responses, distinct differences are observed in the influence patterns of oscillation amplitude and frequency on both porosity and maximum pore size. Specifically, the oscillation amplitude exhibited a similar significance of effect on the two pore defect metrics. In contrast, the oscillation frequency demonstrated a large significance of effect on overall porosity than on the maximum pore size. This suggests a potential optimization strategy for pore removal: first, increasing the oscillation

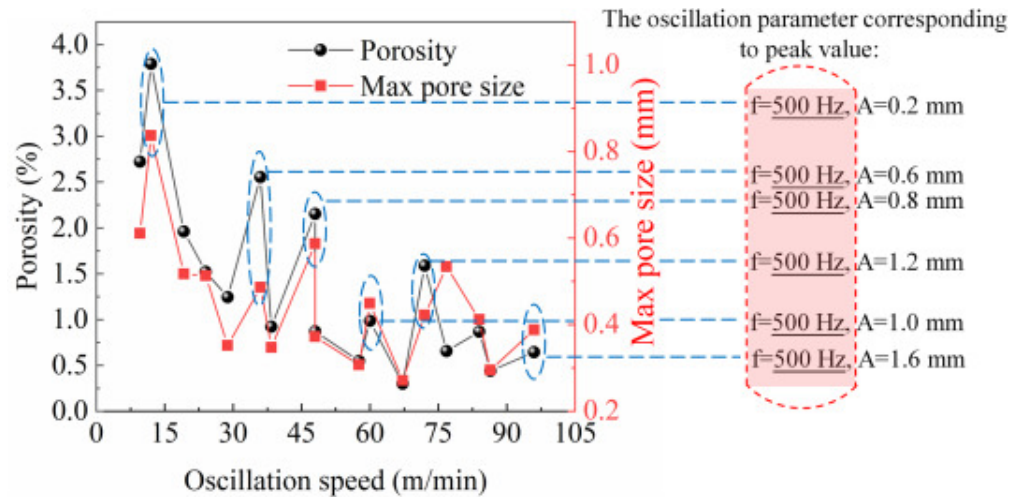
frequency at a fixed amplitude to rapidly reduce overall porosity, and subsequently changing the amplitude to further minimize the maximum pore size. This two-stage approach may lead to a more efficient optimization.

To comprehensively analyze the influence of beam oscillation amplitude and frequency on pore defects, a defined parameter, the beam scanning average velocity V_o , is proposed. This parameter characterizes the average trajectory length of the beam's periodic oscillatory motion along the weld direction per unit time. It is jointly determined by the oscillation amplitude (A) and frequency (f), and its calculation formula is shown as Eq. 13:

$$V_o = \frac{60 f A}{1000} \times 2 \quad (13)$$

By establishing a multiplicative relationship between amplitude and frequency, this formula reflects the combined intensity of the molten pool agitation induced by the beam oscillation.

The statistical results from Fig. 15 reveal that as the oscillation velocity increases, both the porosity and the maximum pore size exhibit an initial decrease followed by a gradual increase. The minimum values for both metrics are observed at an oscillation velocity of 67.2m/min. This trend aligns more closely with the data obtained under a fixed amplitude and varying frequency, suggesting that the oscillation frequency exerts a more dominant influence on porosity and pore size than the amplitude. This conclusion is in close agreement with the trend indicated by the gradients from the aforementioned Bayesian statistical posterior sampling.



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Fig. 15. Effect of average oscillation velocity on porosity and maximum pore size.

In addition, observations indicated that when the scanning speeds were similar, some points exhibited a significant increase in both porosity and maximum pore size. These points share a common characteristic: although their scanning speeds are close to those of the adjacent points, their scanning frequency is noticeably higher than that of the adjacent points. Particularly under the conditions of 0.2mm oscillation amplitude and 500Hz oscillation frequency, both the porosity and the maximum pore size reach their maximum values across the entire experimental parameter range. If the oscillation frequency is too high, it may cause severe keyhole oscillations and frequent collapse. Additionally, energy accumulation occurs at the turning points of the reciprocating motion. Combined with high-frequency impacts, this can easily rupture the liquid film, triggering micro-explosions and spatter. The high-speed imaging results in Fig. 13(d) also reveal significant spatter under these parameters, indicating an unstable melt pool. In contrast, as shown in Fig. 13(c), when the oscillation frequency remains at 500Hz but the oscillation amplitude is increased to 1.6mm, the oscillation-induced spatter is markedly

reduced, and the melt pool becomes more stable. This improvement arises because, at a constant oscillation frequency, increasing the amplitude reduces the effective energy density delivered to each point along the welding path, resulting in a smoother energy input that promotes melt pool stability and mitigates spatter and porosity defects. These findings collectively indicate that, at the same scanning speed, appropriately reducing the oscillation frequency while increasing the oscillation amplitude may represent a viable strategy for achieving lower porosity and a smaller maximum pore size.

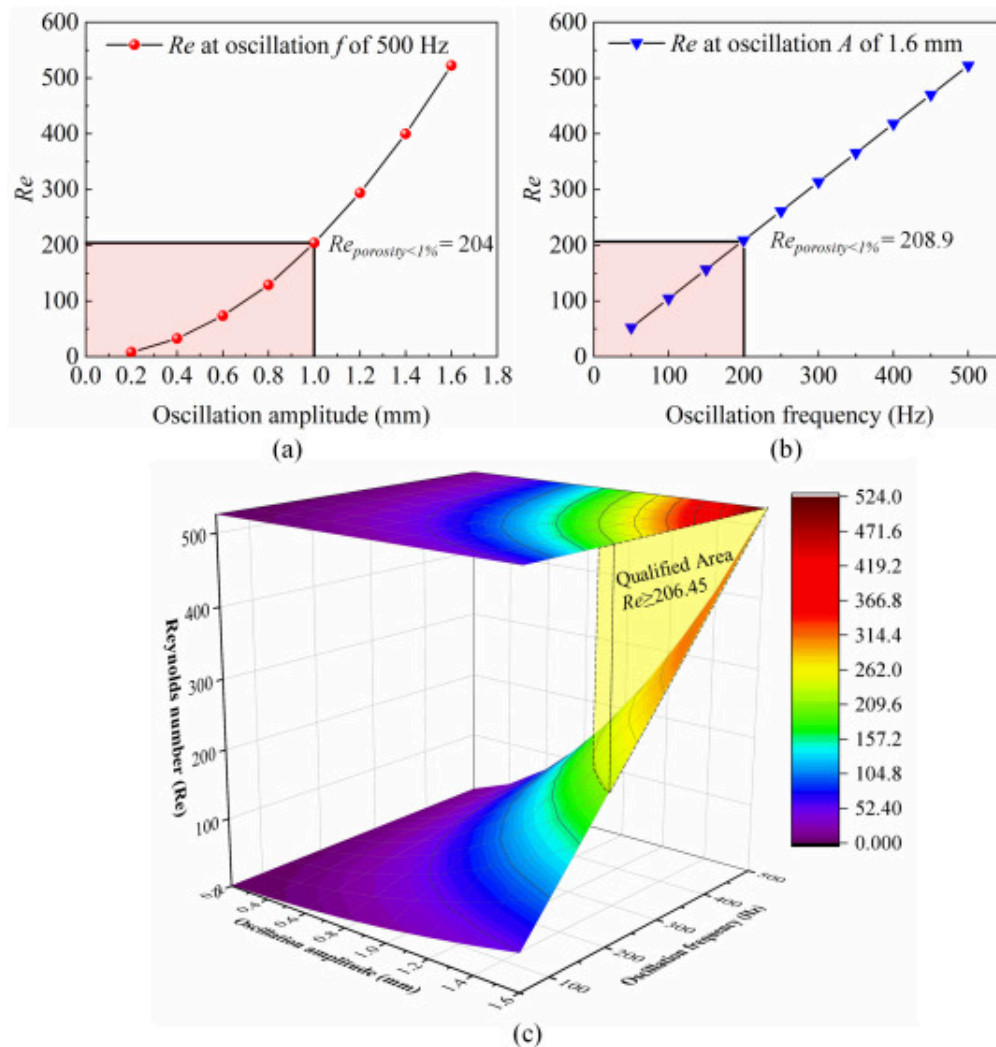
Furthermore, laser oscillation generates intense stirring within the melt pool. An appropriate stirring intensity can promote the interaction between pores and the keyhole, thereby enhancing the suppression of pore defects. The intensity of this melt pool stirring induced by laser beam oscillation is represented by the stirring Reynolds number (Re) in the Eq. 14 [48]:

$$Re = \frac{\rho D v^*}{\mu} \quad (14)$$

where ρ is the melt density, which is 10,200 kg/m³ for Mo, D is the stirring length, and V is the stirring velocity, and μ is the melt viscosity, which is 0.05 Pa.s for Mo. In laser oscillation welding, the keyhole, which acts as the stirring apparatus, justifies the substitution of D and v^* with A (oscillation amplitude) and V_k (the average keyhole velocity), respectively. Substituting A and V_k (where $V_k = 2Af$) into Eq. 14 get Eq. 15.

$$Re = \frac{\rho D v^*}{\mu} = \frac{2f\rho A^2}{\mu} \quad (15)$$

The Reynolds numbers corresponding to the different oscillation parameters in Table 2 and Table 3 were calculated using this formula, with the results presented in Fig. 16(a), (b). Typically, a Reynolds number greater than 1000 induces turbulent flow, while a value less than 1000 results in laminar or transitional flow status [49]. In the parameter range investigated in this study, the melt pool consistently exhibited a transitional flow state, indicating that the selected parameters were reasonable.



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Fig. 16. Reynolds number (Re) under different oscillation parameters. (a) Re under different oscillation amplitudes at a fixed frequency; (b) Re under different oscillation frequencies at a fixed amplitude; (c) Response surface diagram of Reynolds number corresponding to oscillation frequency and oscillation amplitude.

According to the ISO 13919 standard, welded joints with a porosity level below 1% meet the requirements for industrial applications. This also satisfies the acceptance criteria for a Class I weld joint as specified in the JB/T 4730.2 standard. As can be seen from Fig. 16(a) and Fig. 16(b), the critical Reynolds numbers corresponding to a porosity of less than 1% are 204 and 208.9 for the fixed oscillation amplitude and frequency conditions, respectively. These two values are very close, indicating that for the $\Phi 19 \times 0.9$ mm Mo alloy pipe in this study, the average minimum threshold of the Reynolds number is approximately 206.45. The Reynolds numbers under the coupling effect of different oscillation amplitudes and frequencies within the experimental parameter range were calculated, and the corresponding 3D response surface was established. Combined with the mean value of boundary thresholds, the feasible parameter region under the interaction of amplitude and frequency was preliminarily estimated, as indicated by the yellow area in Fig. 16 (c). Theoretically, a weld joint with porosity below 1% can be achieved when the Reynolds number exceeds this value. However, it is noteworthy that porosity shows an increasing trend with a further rise in oscillation frequency. This suggests that the Reynolds number alone cannot be the sole consideration; a comprehensive evaluation with specific oscillation parameters is necessary. Furthermore, two groups of verification tests were carried out on Mo-5Re alloy cladding with an outer diameter of 9.5 mm and a wall thickness of 0.8 mm using process parameters exceeding the Reynolds number threshold. The porosity still met the qualification standard. Nevertheless, when the parameters beyond the threshold were adopted for welding the socket joint assembled by 19 mm outer-diameter Mo alloy pipe and end plug in this work, the porosity failed to reach the qualified level. Detailed specimen dimensions and test results are provided in the supplementary information. Preliminary limited verification indicated that the Reynolds number threshold determined in this study is possibly insensitive to the size of Mo alloy tubes, but highly sensitive to joint configuration. Nevertheless, the Reynolds number remains a valuable reference for optimizing the oscillation process.

Furthermore, within the range of welding and oscillation parameters employed in this study, a threshold effect of oscillation frequency on porosity control is observed, whereas no optimal peak for oscillation amplitude has been reached yet. Given this phenomenon, further increasing the oscillation amplitude at the optimal oscillation frequency may help to further reduce pore defects.

4. Conclusion

This study systematically investigates the influence of laser longitudinal beam oscillation welding on pore defects in weld joints of Mo alloy pipes. The main conclusions are as follows:

- 1) Increasing the longitudinal oscillation amplitude and frequency within a certain range can effectively reduce the porosity and maximum pore size. However, when the oscillation frequency exceeds 350Hz, both metrics increase. At similar oscillating speeds, excessively high frequencies tend to result in higher porosity and larger maximum pore sizes.
- 2) Oscillation frequency and amplitude do not affect the spatial distribution or size distribution of pores. Both porosity and maximum pore size initially increase and then decrease from the inner wall to the outer wall of the Mo alloy pipe, peaking at a distance of 0.5–0.7mm from the inner wall. A significant number of visible pores accumulate near the fusion line, and the number of pores increases as their size decreases.
- 3) From a statistical perspective using Bayesian methods, there is more than 95% confidence that increasing the oscillation frequency has a significantly greater effect on reducing the porosity than on decreasing the maximum pore size. In contrast, the influence of oscillation amplitude on both metrics is statistically comparable under small sample size.

- 4) Longitudinal beam oscillation results in a more uniform energy distribution and a slender molten pool. However, high oscillation frequencies are detrimental to molten pool stability. Under the parameters and size of the heat pipe studied, welds with acceptable porosity are achieved when the Reynolds number exceeds about 206.45.

Based on these findings, propose a possible efficient method for controlling pores, the oscillation amplitude and frequency can first be preset according to the Reynolds number threshold. Prioritizing an increase in oscillation frequency to suppress the porosity, followed by adjusting the amplitude to further control the maximum pore size, thereby achieving efficient management of pore defects. Nevertheless, the spatial distribution of pores—particularly near the fusion line—remains difficult to completely remove through changing oscillation parameters and represents a key issue to be addressed in future research.

CRediT authorship contribution statement

Xujing Zhang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Linjie Zhang:** Writing – review & editing, Supervision, Funding acquisition. **Jiaxuan Zhao:** Validation, Methodology, Data curation. **Jie Ning:** Writing – review & editing. **Jian Long:** Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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Supplementary material

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